

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

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BÁO CÁO BÀI TẬP

BTVN 02

Họ Và Tên: Nguyễn Hoàng Long

Mã Số Sinh Viên: 20235367

Học Phần: IT4060

Mã Lớp: 168503

B01 :CHAT_SERVER

Ảnh minh họa khởi chạy server, khởi chạy client:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc chat_server.c
-o chat_server
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./chat_server
Server is listening on port 8080...
New client connected: 4
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc client.c -o client
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui long nhap ten (id: name):
█
```

Ảnh minh họa client gửi sai cú pháp:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc chat_server.c
-o chat_server
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./chat_server
Server is listening on port 8080...
New client connected: 4
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc client.c -o client
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui long nhap ten (id: name):
yusjnddad djdsjds
Received: Sai cu phap. Nhap lai (id: name):
█
```

Ảnh minh họa client gửi đúng dữ liệu:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc chat_server.c
-o chat_server
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./chat_server
Server is listening on port 8080...
New client connected: 4
Client 4 xác thực thành công với ID: y25
```

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc client.c -o client
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui lòng nhập tên (id: name):
yusjnddad djdsjds
Received: Sai cú pháp. Nhập lại (id: name):
y25: yuya
Received: Xác thực thành công! Bạn đã vào phòng.
```

Ảnh minh họa client gửi tin nhắn đến các client khác:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc chat_server.c
-o chat_server
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./chat_server
Server is listening on port 8080...
New client connected: 4
Client 4 xác thực thành công với ID: y25
New client connected: 5
Client 5 xác thực thành công với ID: S25
Received from 4: hello ervery body
Received from 5: hi y25
```

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc client.c -o client
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui lòng nhập tên (id: name):
yusjnddad djdsjds
Received: Sai cú pháp. Nhập lại (id: name):
y25: yuya
Received: Xác thực thành công! Bạn đã vào phòng.

hello ervery body
Received: S25: hi y25
```

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui lòng nhập tên (id: name):
S25: minh
Received: Xác thực thành công! Bạn đã vào phòng.

Received: y25: hello ervery body

hi y25
```

Ảnh minh họa client ngắt kết nối với server:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc chat_server.c
-o chat_server
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./chat_server
Server is listening on port 8080...
New client connected: 4
Client 4 xác thực thành công với ID: y25
New client connected: 5
Client 5 xác thực thành công với ID: S25
Received from 4: hello ervery body
Received from 5: hi y25
Client 4 disconnected
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc client.c -o client
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui long nhap ten (id: name):
yusjnddad djdsjds
Received: Sai cu phap. Nhap lai (id: name):
y25: yuya
Received: Xac thuc thanh cong! Ban da vao phong.

hello ervery body
Received: S25: hi y25

exit
Exiting...
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ █

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client
Connected to server on port 8080...
Received: Vui long nhap ten (id: name):
S25: minh
Received: Xac thuc thanh cong! Ban da vao phong.

Received: y25: hello ervery body

hi y25
```

B02: TELNET_SERVER

Ảnh file acc.txt (lưu thông tin tài khoản):

```
acc.txt  U X  out.txt  U
BTVN: 20235367 > btvn 2 > acc.txt
1  yuya 123456
2  admin 123456
3
```

Ảnh minh họa khởi chạy server:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc telnet_server.  
c -o telnet_server  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./telnet_server  
Server is listening on port 8080...
```

Ảnh minh họa khởi chạy client:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc telnet_server.  
c -o telnet_server  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./telnet_server  
Server is listening on port 8080...  
New client connected: 4  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client  
Connected to server on port 8080...  
Received: Enter user and pass:
```

Ảnh minh họa client đăng nhập sai:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc telnet_server.  
c -o telnet_server  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./telnet_server  
Server is listening on port 8080...  
New client connected: 4  
█
```

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client  
Connected to server on port 8080...  
Received: Enter user and pass:  
yuya 12hdhsj  
Received: Authentication failed! Please try again.  
█
```

Ảnh minh họa client đăng nhập thành công:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc telnet_server.  
c -o telnet_server  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./telnet_server  
Server is listening on port 8080...  
New client connected: 4  
█
```

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client  
Connected to server on port 8080...  
Received: Enter user and pass:  
yuya 12hdhsj  
Received: Authentication failed! Please try again.  
  
yuya 123456  
Received: Authentication successful!  
█
```

Ảnh minh họa client gửi lệnh đến server:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc telnet_server.  
c -o telnet_server  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./telnet_server  
Server is listening on port 8080...  
New client connected: 4  
Command from 4: dir  
█
```

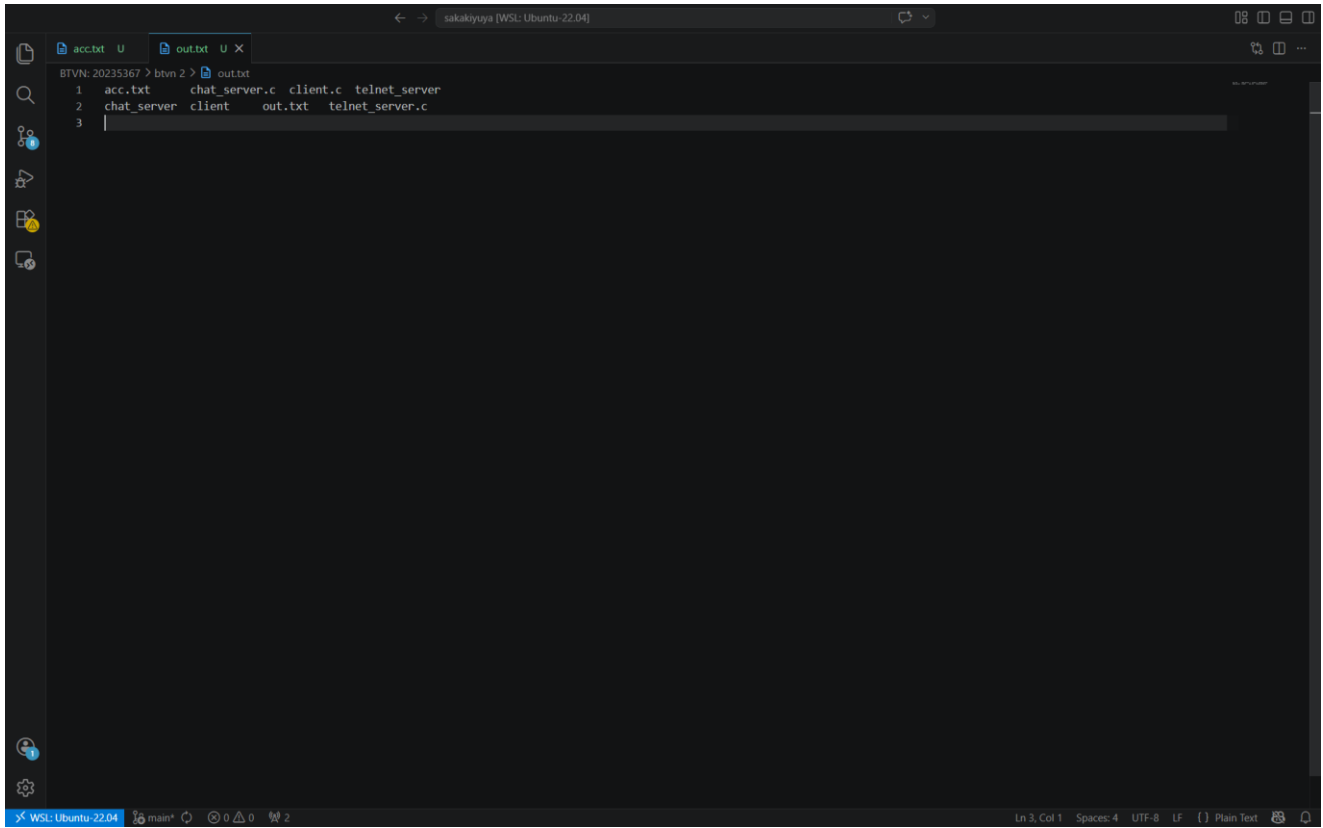
```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client  
Connected to server on port 8080...  
Received: Enter user and pass:  
yuya 12hdhsj  
Received: Authentication failed! Please try again.  
  
yuya 123456  
Received: Authentication successful!  
  
dir  
Received: Command output:  
acc.txt      chat_server.c  client.c  telnet_server  
chat_server  client        out.txt   telnet_server.c  
█
```

Ảnh minh họa client ngắt kết nối đến server:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ gcc telnet_server.  
c -o telnet_server  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./telnet_server  
Server is listening on port 8080...  
New client connected: 4  
Command from 4: dir  
Client 4 disconnected  
█
```

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ ./client  
Connected to server on port 8080...  
Received: Enter user and pass:  
yuya 12hdhsj  
Received: Authentication failed! Please try again.  
  
yuya 123456  
Received: Authentication successful!  
  
dir  
Received: Command output:  
acc.txt      chat_server.c  client.c  telnet_server  
chat_server  client        out.txt   telnet_server.c  
  
exit  
Exiting...  
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 2$ █
```

Ảnh file out.txt sau khi chạy xong:



The image shows a screenshot of a Visual Studio Code editor window. The title bar at the top indicates the file is 'sakakiyaya [WSL: Ubuntu-22.04]'. The editor has two tabs open: 'acc.txt' and 'out.txt'. The 'out.txt' tab is active, showing the following content:

```
BTVN: 20235367 > btvn 2 > out.txt
1 acc.txt chat_server.c client.c telnet_server
2 chat_server client out.txt telnet_server.c
3
```

The status bar at the bottom shows 'WSL: Ubuntu-22.04', 'main', and 'Ln 3, Col 1 | Spaces: 4 | UTF-8 | LF | Plain Text'.